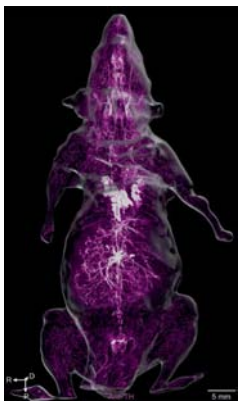
Neuroscience

ViSOR puts the periphery in focus



Labelling of sympathetic nerves using ARCHmap-blockface-ViSOR. Reprinted from M.-Y. Shi et al. *Cell* 188, 3897–3915 (2025), CCBY 4.0.

Tissue clearing and optical imaging are frequently used techniques in connectomics and have been applied successfully to image and map individual neurons in the mammalian brain. Imaging the mouse peripheral nervous system (PNS) at resolutions high enough to differentiate individual neurons, however, has proven challenging owing to its size, complexity, and loss of resolution when imaging through large volumes.

Researchers from the University of Science and Technology of China led by Guo-Qiang Bi demonstrate an approach capable of imaging the mouse PNS at micrometer resolutions, based on their published ViSOR method. In ViSOR, cleared, fluorescently labeled tissue is imaged by rapidly scanning a light beam and perpendicular objective lens over the sample surface at a 45° angle.

The authors developed a custom tissue clearing step (ARCHmap) involving cardiac perfusion of an aqueous clearing solution, CUBIC-LH, to maximize its permeation through the mouse body. The cleared body is then embedded in hydrogel using a protocol optimized for mechanical strength. Using a Triton X-100-based refractive index matching solution, the mouse is imaged using a ViSOR platform configured for blockface imaging using a micrometer-precision sequential cutting and imaging approach. Images are then reconstructed computationally to produce a full 3D volume.

Reconstructed volumes from mice labelled with neuronal markers showed several features of the PNS not previously resolved by other whole-body imaging techniques, including individual fascicles — bundles of nerve fibers — in cranial nerves, including ocular nerves, and the presence of mesh-like sympathetic nerves around vasculature throughout the head, torso, limbs and internal organs. Tracing vagus nerve projections showed an often direct one-to-one connection of neurons from the dorsal motor nucleus in the medulla to individual organs. Imaging neurons from the spine demonstrated that the length and degree of branching of their projections varies, depending on their position along the spinal column and their motor or sensory identity. Further studies seem likely to identify additional fine details of the PNS.

Future work on the method will aim to further optimize the tissue clearing protocol and adapt the protocol to allow restaining and analysis of tissue sections collected during the blockface approach.

Joseph Willson
Nature Methods

Original reference: *Cell* **188**, 3897–3915 (2025)

Bioinformatics

Benchmarking genomic language models

Supervised deep learning models have a dazzling track record in many computational genomics tasks, but their success relies on vast (and often costly) experimental data for training. Recently, genomic language models (gLMs), whose pretraining needs only (although large numbers of) DNA sequences, have manifested as a potentially appealing alternative and are interesting many researchers in the genomics and computational biology community, including Peter Koo of Cold Spring Harbor Laboratory. "We were initially excited by the growing class of gLMs that aim to learn unsupervised representations of DNA," he says. However, after building these models with his team, "We found that they consistently underperformed well-established supervised models." Intrigued to know whether these observations prevail more generally, Koo and his colleagues shifted their project to perform a rigorous evaluation of gLMs.

Challenges abound for benchmarking analysis in such a fast-paced area. Although new gLMs keep emerging, issues with code and data availability often hinder full reproducibility. "Many functional genomics modeling papers provided code and data, but these were often incomplete or difficult to adapt," says Koo. This led the team to concentrate on a small but representative set of gLMs whose data and model baselines could be reliably obtained. Another important distinction of their benchmarking study compared to previous efforts is the tasks they designed. "The key innovation of our evaluation is its focus on biologically aligned tasks that are tied to open questions in gene regulation," notes Koo. "In contrast, most existing benchmarks rely on classification tasks that originated in the machine learning literature and continue to be propagated in gLM studies, despite being disconnected from how models would be used to advance biological understanding and discovery."

The key results, as revealed by multiple tasks and comparison with both simple and supervised learning models, are that current gLMs, even when pretrained on entire genomes, do not appear to learn the *cis*-regulatory logic required for meaningful prediction of gene regulation, notes Koo. "The excitement surrounding gLMs and the belief that scale alone might be sufficient had created the expectation that these models could uncover more than they currently do. Our findings suggest that simply increasing the model scale or the number of training tokens is not the key to progress, especially given that much of the genome is not functionally informative."

Instead of dismissing the potential of gLMs, Koo believes their work will help calibrate expectations. "By identifying current limitations, we hope to guide future efforts toward models that are both scientifically useful and biologically grounded." In his view, promising directions to pursue include rethinking the training objectives, developing biologically tailored models, and better identifying applications where gLMs are most likely to offer a clear advantage.

Lin Tang
Nature Methods

Original reference: *Genome Biol.* **26**, 203 (2025)